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Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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Welcome to the course **Fundamentals and Introduction to Machine Learning** designed for students of the Faculties of Sciences and Chemical Sciences at the University of Salamanca.

Purpose of the Book

This book guides you on a journey from classical statistical concepts to modern deep learning models. The goal is for you to **understand how machines learn from data** and for you to be able to build, train, and evaluate models that solve real-world problems.

Content Structure

The book is divided into three main sections:

Section 1: Fundamentals

Understand the paradigm shift: from fixed-rule programming to data-driven learning.

- Historical origins of AI and machine learning
- Fundamental concepts: data, features, labels
- Evaluation metrics and cross-validation

Section 2: Supervised Learning

Learn to model continuous variables (regression) and categorical variables (classification).

- Linear and polynomial regression
- Linear and non-linear classifiers
- Neural networks

Section 3: Unsupervised Learning

Discover hidden patterns in unlabeled data.

- Clustering (grouping)
- Dimensionality reduction
- Anomaly detection

Who Is This For?

This course is designed for:

- Undergraduate students in Science, Physics, Statistics, Computer Science, or Engineering
- Anyone interested in machine learning and data analysis
- Professionals looking to update their skills

Prerequisites

While the book is self-contained, it helps to have:

- Basic Python knowledge or willingness to learn
- Familiarity with linear algebra and calculus (but not required)
- Analytical thinking and patience with mathematical concepts

How to Use This Book

Each chapter includes:

- **Theoretical concepts** explained clearly
- **Practical examples** with executable Python code
- **Visualizations** to better understand results
- **Exercises** to reinforce learning

It's recommended to follow the established order, though you can consult specific sections as needed.

Let's begin! Go to Section 1: Fundamentals to start your machine learning journey.

Part I

Section 1 - Fundamentals

SECTION 1: FUNDAMENTALS

Welcome to the first section of the course. Here we will understand the basic principles of machine learning and how it differs from the traditional programming approach.

1.1 Why Learn Machine Learning?

In traditional programming, a programmer explicitly writes all the rules that a program must follow:

```
if temperature > 30:
    turn_on_air_conditioning()
if temperature < 15:
    turn_on_heating()
```

But many real-world problems are too complex to describe with fixed rules. For example:

- How do we automatically distinguish a cat from a dog in a photograph?
- How do we predict if a bank transaction is fraudulent?
- How do we classify emails as spam or legitimate?

Machine learning inverts the paradigm: instead of writing the rules, we teach the program to *learn* the rules from examples.

1.2 What Will You Find in This Section?

1. **History and Evolution:** We'll learn how AI was born and evolved to reach the modern models we use today.
2. **Fundamental Concepts:** We'll learn the essential technical vocabulary for working with machine learning models.

These foundations are critical: without a solid understanding of these concepts, it will be difficult to understand the more advanced techniques in the later sections.

Let's start: Go to Chapter 1.1 to explore the fascinating history of machine learning.

1.1 HISTORY AND EVOLUTION OF MACHINE LEARNING

(Content translated from Spanish version - Full English version coming soon)

This chapter covers the fascinating history of AI and ML from Turing's foundational question through the recent era of Large Language Models. We discuss:

- The 1950s origins with Alan Turing and McCulloch-Pitts neurons
- The perceptron and early optimism in connectionism
- The AI winter and symbolic AI limitations
- The resurgence with statistical learning approaches
- The deep learning revolution of the 2010s
- The modern era of Transformers and LLMs

2.1 Key Takeaways

Machine learning has gone through cycles of hype and disenchantment, but steady progress has been driven by three factors: more data, more computational power, and better algorithms.

Next: Chapter 1.2 covers Fundamental Concepts needed to work with any ML model.

1.2 FUNDAMENTAL CONCEPTS

(Content translated from Spanish version - Full English version coming soon)

This chapter introduces the essential vocabulary and concepts for working with machine learning models:

- **Data, Features, and Labels:** The basic structure of datasets
- **Overfitting and Underfitting:** The bias-variance tradeoff
- **Cross-Validation:** Rigorous evaluation without bias
- **Evaluation Metrics:** Precision, recall, F1, AUC for classification; MSE, MAE, RMSE for regression
- **The ML Pipeline:** From data collection through deployment

Next: Section 2 covers Supervised Learning approaches.

Part II

Section 2 - Supervised Learning

SECTION 2: SUPERVISED LEARNING

Welcome to the second section of the course. Here we'll learn to build models that learn from labeled examples.

4.1 What is Supervised Learning?

In supervised learning, we have:

- **Data:** input features (X)
- **Labels:** correct answers (y)
- **Goal:** learn the relationship $X \rightarrow y$

It's like learning with a "supervisor" who tells you if your answers are correct.

4.2 Two Types of Problems

4.2.1 1. Regression

We predict a **continuous value** (a real number):

- Predict house price
- Predict tomorrow's temperature
- Predict rainfall amount
- Predict population growth

4.2.2 2. Classification

We predict a **category** or **class**:

- Diagnose if a person has a disease (yes/no)
- Classify emails as spam or legitimate
- Recognize if an image is a cat, dog, or bird
- Predict if a customer will churn

4.3 What You'll Find in This Section

4.3.1 Chapter 2.1: Regression

- Linear Regression: the simplest and most fundamental model
- Ordinary Least Squares: how regression trains
- Gradient Descent: the optimization algorithm
- Polynomial Regression and Regularization: more complex models

4.3.2 Chapter 2.2: Classification

- Logistic Regression: binary classification
- Confusion Matrix and Metrics: evaluating classifiers
- Decision Trees and Random Forests: interpretable models
- Support Vector Machines: optimal separation
- Ensemble Methods: combining multiple models
- Neural Networks: introduction to deep learning

4.4 Recommended Structure

It's recommended to learn in this order:

1. **Linear Regression** → understand fundamentals
2. **Gradient Descent** → understand how models learn
3. **Polynomial Regression** → handle non-linearity
4. **Logistic Regression** → extend regression to classification
5. **Classification Algorithms** → learn multiple approaches
6. **Ensemble Methods** → combine model strengths
7. **Neural Networks** → the future of machine learning

4.5 Learning Objectives

After completing this section, you'll be able to:

- ✓ Implement linear regression from scratch
- ✓ Understand how gradient descent works
- ✓ Apply regularization to prevent overfitting
- ✓ Choose the right algorithm for your problem
- ✓ Evaluate classification models correctly
- ✓ Build a simple neural network

Let's start: Go to Chapter 2.1 to learn about Regression, the foundation of many ML models.

2.1 REGRESSION

(Content translated from Spanish version - Full English version coming soon)

This chapter covers regression, predicting continuous values:

- **Linear Regression:** the simplest model
- **Ordinary Least Squares (OLS):** exact solution for linear regression
- **Gradient Descent:** general optimization algorithm for any model
- **Polynomial Regression:** capturing non-linear relationships
- **Regularization (Ridge, Lasso):** preventing overfitting

Next: Chapter 2.2 covers Classification approaches.

2.2 CLASSIFICATION

(Content translated from Spanish version - Full English version coming soon)

This chapter covers classification, predicting categorical variables:

- **Logistic Regression:** binary classification with sigmoid
- **Confusion Matrix:** visualizing classifier performance
- **Metrics:** Precision, Recall, F1-Score, AUC-ROC
- **Decision Trees:** interpretable but prone to overfitting
- **Random Forest:** ensemble of trees
- **Support Vector Machines:** optimal hyperplane separation
- **Gradient Boosting:** sequential error correction (XGBoost, LightGBM)
- **Neural Networks:** introduction to deep learning

Next: Section 3 covers Unsupervised Learning approaches.

Part III

Section 3 - Unsupervised Learning

SECTION 3: UNSUPERVISED LEARNING

Welcome to the third and final main section of the course. Here we'll explore techniques to extract patterns from **unlabeled data**.

7.1 What is Unsupervised Learning?

In unsupervised learning:

- We have **only features** (X)
- We **don't have labels** (no y)
- We want to **discover hidden patterns**

It's like exploring unknown territory without a map. The machine must find its own structure.

7.2 Three Main Types

7.2.1 1. Clustering

Grouping similar data:

- Segment customers by behavior
- Find communities in social networks
- Classify documents by topic
- Identify cancer cells in images

7.2.2 2. Dimensionality Reduction

Simplifying complex data:

- Visualizing high-dimensional data
- Removing noise
- Accelerating later algorithms
- Interpretable understanding

7.2.3 3. Anomaly Detection

Identifying rare or unusual cases:

- Fraud in transactions
- System failures
- Unusual user behavior
- Scientific data outliers

7.3 What You'll Find in This Section

7.3.1 Chapter 3.1: Clustering

- **K-Means**: partition data into groups
- **Elbow Method**: select optimal number of clusters
- **Hierarchical Clustering**: dendrograms and merging
- **DBSCAN**: density-based clustering

7.3.2 Chapter 3.2: Dimensionality Reduction

- **PCA (Principal Component Analysis)**: maximize explained variance
- **t-SNE and UMAP**: non-linear visualization

7.3.3 Chapter 3.3: Anomaly Detection

- **Identifying unusual patterns**: what is an anomaly
- **Detection methods**: practical applications

7.4 Main Challenge: Evaluation

Without labels, how do we know our results are correct?

```
Supervised Learning:  
Data → Model → Prediction → Compare with y → Error  
  
Unsupervised Learning:  
Data → Model → Structure ??? → Is it correct?
```

This is more challenging and requires:

- **Domain intuition**: Does the result make sense?
- **Internal metrics**: silhouette, Davies-Bouldin index
- **External validation**: labels available later (if any)

7.5 Learning Objectives

After completing this section, you'll be able to:

- ✓ Implement K-Means from scratch
- ✓ Choose the optimal number of clusters
- ✓ Visualize high-dimensional data
- ✓ Detect anomalies in real data
- ✓ Understand the limitations of each technique

Let's start: Go to Chapter 3.1 to learn about Clustering, the most popular unsupervised approach.

3.1 CLUSTERING

(Content translated from Spanish version - Full English version coming soon)

This chapter covers clustering techniques for grouping similar data:

- **K-Means:** partition into K clusters, minimizing within-cluster distance
- **Elbow Method:** choosing optimal K
- **Hierarchical Clustering:** dendrograms and linkage methods
- **DBSCAN:** density-based clustering, arbitrary shapes

Next: Chapter 3.2 covers Dimensionality Reduction.

3.2 DIMENSIONALITY REDUCTION

(Content translated from Spanish version - Full English version coming soon)

This chapter covers techniques to reduce data complexity while preserving essential information:

- **PCA (Principal Component Analysis):** maximize variance explained
- **t-SNE:** visualize complex relationships in 2D
- **UMAP:** balance between PCA and t-SNE
- **Curse of Dimensionality:** why reduction matters
- **Applications:** compression, visualization, preprocessing

Next: Chapter 3.3 covers Anomaly Detection.

3.3 ANOMALY DETECTION

(Content translated from Spanish version - Full English version coming soon)

This chapter covers techniques for identifying rare and unusual patterns:

- **Fundamental Concepts:** point, contextual, collective anomalies
- **Statistical Methods:** Gaussian univariate approach
- **Distance-based Methods:** k-NN, LOF
- **Isolation Forest:** efficient tree-based approach
- **Autoencoders:** neural network approach
- **Evaluation Metrics:** Recall more important than Accuracy
- **Applications:** fraud detection, preventive maintenance, intrusion detection

Congratulations! You've completed the course on Fundamentals and Introduction to Machine Learning. You now have solid foundations to explore more advanced topics.